

Chapter 3

Continuity

§ Section: 2.5

3.1 Continuity at a point

The limit of a function as x approaches a can often be found simply by calculating the value of the function at a . Functions with this property are called *continuous* at a .

Definition:

A function f is **continuous at a number** $x = a$ if

$$\lim_{x \rightarrow a} f(x) = f(a)$$

Remark:

If $f(x)$ is continuous at $x = a$, then

1. $f(a)$ is defined (that is, a is in the domain of $f(x)$).
2. $\lim_{x \rightarrow a} f(x)$ exists.
3. $\lim_{x \rightarrow a} f(x) = f(a)$.

Roughly speaking, the graph of a continuous function has no hole or break.



Continuity means nice
and smooth!



You can draw it without
raising your pen



$$\lim_{x \rightarrow a} f(x) = f(a)$$



$$\forall \epsilon > 0 \exists \delta > 0 \text{ s.t.}$$

$$|x - a| < \delta \Rightarrow |f(x) - f(a)| < \epsilon$$



The pre-image of an
open set is open

Example 1.1

Find constant c such that $f(x)$ is continuous at $x = 3$, where

$$f(x) = \begin{cases} [x] + cx + 4, & \text{for } x \geq 3, \\ \frac{|-x^2 + x + 6|}{x - 3}, & \text{for } x < 3. \end{cases}$$

Sol:

$\therefore f(x)$ is contz at $x=3$

$\therefore \lim_{x \rightarrow 3} f(x)$ exists and $\lim_{x \rightarrow 3} f(x) = f(3)$.

Hence $\lim_{x \rightarrow 3^+} f(x) = f(3)$. $\left(\lim_{x \rightarrow 3^+} f(x) = \lim_{x \rightarrow 3^+} [x] + cx + 4 = 3 + 3c + 4 \right)$

$$\lim_{x \rightarrow 3^-} f(x) = f(3) = [3] + 3c + 4 = 7 + 3c$$

$$\lim_{x \rightarrow 3^-} \frac{|x-3| |x+2|}{x-3} = \lim_{x \rightarrow 3^-} \frac{(3-x)(x+2)}{x-3} = -5.$$

$$\Rightarrow 7 + 3c = -5 \Rightarrow c = -4.$$

3.2 Continuity on an Interval and Discontinuity

Definition:

A function f is **continuous from the right** at a number a if

$$\lim_{x \rightarrow a^+} f(x) = f(a)$$

and f is **continuous from the left** at a if

$$\lim_{x \rightarrow a^-} f(x) = f(a)$$

Definition:

A function f is **continuous on an interval** if it is continuous at every number in the interval. (If f is defined only on one side of an endpoint of the interval, we understand *continuous* at the endpoint to mean *continuous from the right* or *continuous from the left*.)

Definition:

If f is defined near a (i.e. f is defined on an open interval containing a , **except perhaps at** a), we say that f is **discontinuous at a** (or f has a **discontinuity at a**) if f is not continuous at a .

Type	Definition	Examples
Removable Discontinuity	$\lim_{x \rightarrow a} f(x)$ exists but $f(a)$ is not defined or $\lim_{x \rightarrow a} f(x) \neq f(a)$.	$f(x) = \frac{x^2-1}{x-1}$ at $x=1$.
Jump Discontinuity	$\lim_{x \rightarrow a^-} f(x)$ and $\lim_{x \rightarrow a^+} f(x)$ exist, but $\lim_{x \rightarrow a^-} f(x) \neq \lim_{x \rightarrow a^+} f(x)$.	$f(x) = [x]$ at $x=n \in \mathbb{N}$.
Infinite Discontinuity	$\lim_{x \rightarrow a} f(x) = \pm\infty$.	$f(x) = \frac{1}{1-x}$ at $x=1$.
Essential Discontinuity	Near $x = a$, $f(x)$ varies rapidly.	$f(x) = \cos(\frac{1}{x})$ at $x=0$.

3.3 Some Operators That Preserve Continuity

Theorem:

If f and g are continuous at a and c is a constant, then the following functions are also continuous at a :

1. $f + g$
2. $f - g$
3. cf
4. fg
5. $\frac{f}{g}$ if $g(a) \neq 0$

Theorem:

If f is continuous at b and $\lim_{x \rightarrow a} g(x) = b$, then $\lim_{x \rightarrow a} f(g(x)) = f(b)$. In other words,

$$\lim_{x \rightarrow a} f(g(x)) = f\left(\lim_{x \rightarrow a} g(x)\right)$$

Remark:

The above theorem can be proved by the precise definition of a limit. The continuity of f is necessary. In fact, there are examples such that $\lim_{x \rightarrow a} g(x) = b$ and $\lim_{x \rightarrow b} f(x) = L$, but $\lim_{x \rightarrow a} f(g(x)) \neq L$.

Theorem:

If g is continuous at a and f is continuous at $g(a)$, then the composite function $f \circ g$ given by $(f \circ g)(x) = f(g(x))$ is continuous at a .

Example 3.1

Show that if $f(x)$ and $g(x)$ are continuous at $x = a$ and c is a constant, then $f(x) \pm g(x)$, $cf(x)$, and $f(x) \cdot g(x)$ are continuous at $x = a$. If we further assume that $g(a) \neq 0$, then $\frac{f(x)}{g(x)}$ is continuous at $x = a$.

sol: $\because f(x)$ and $g(x)$ are cont. at $x=a$.

$$\therefore \lim_{x \rightarrow a} f(x) = f(a), \quad \lim_{x \rightarrow a} g(x) = g(a).$$

$$\text{Then } \lim_{x \rightarrow a} f(x) \cdot g(x) = \lim_{x \rightarrow a} f(x) \cdot \lim_{x \rightarrow a} g(x) = f(a) \cdot g(a)$$

$$\Rightarrow f(x) \cdot g(x) \text{ is cont. at } x=a.$$

Similarly, $f(x) \pm g(x)$, $cf(x)$, $\frac{f(x)}{g(x)}$ are cont. at $x=a$.

Example 3.2

Show that $\lim_{x \rightarrow a} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x \rightarrow a} f(x)}$ if $f(x) \geq 0$ for x near a and $\lim_{x \rightarrow a} f(x)$ exists.

sol: $\because \lim_{x \rightarrow a} \sqrt[n]{x} = \sqrt[n]{a}$ i.e. $g(x) = \sqrt[n]{x}$ is cont. on its domain.

$\therefore$ By the theorem, we have

$$\lim_{x \rightarrow a} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x \rightarrow a} f(x)}.$$

3.4 Examples of Continuous Functions

Theorem:

The following types of functions are continuous at every number in their domains:

polynomials

rational functions

exponential functions

root functions

trigonometric functions

Remark:

The inverse of a one-to-one continuous function is also continuous. Therefore, logarithmic functions, as the inverses of exponential functions, are continuous. Similarly, inverse trigonometric functions are continuous on their respective domains.

Example 4.1

Show that $\sin x$ and $\cos x$ are continuous at $x = 0$.

sol: $\because \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$ (We will prove it later.)

$$\therefore \lim_{x \rightarrow 0} \sin x = \lim_{x \rightarrow 0} \frac{\sin x}{x} \cdot x = 1 \cdot 0 = 0 = \sin 0.$$

$\Rightarrow \sin x$ is cont: at $x=0$.

When x is near 0, $\cos x = \sqrt{1 - (\sin x)^2}$

$$\lim_{x \rightarrow 0} \cos x = \lim_{x \rightarrow 0} \sqrt{1 - \sin^2 x} = \sqrt{\lim_{x \rightarrow 0} (1 - \sin^2 x)} = \sqrt{1 - 0} = 1 = \cos 0.$$

Example 4.2

Show that $\sin x$ and $\cos x$ are continuous at any point $x = x_0$.

$\Rightarrow \cos x$ is cont: at $x=0$.

sol: Prove $\lim_{x \rightarrow x_0} \sin x = \sin x_0 \Leftrightarrow \lim_{h \rightarrow 0} \sin(x_0 + h) = \sin x_0$.

$$\lim_{h \rightarrow 0} \sin(x_0 + h) = \lim_{h \rightarrow 0} \sin x_0 \cos h + \cos x_0 \sin h$$

$$= \sin x_0 \cdot \lim_{h \rightarrow 0} \cos h + \cos x_0 \cdot \lim_{h \rightarrow 0} \sin h = \sin x_0 \cdot 1 + \cos x_0 \cdot 0 = \sin x_0.$$

This proves that $\sin x$ is cont: at $x = x_0$.

Example 4.3

Compute $\lim_{x \rightarrow 1} \sec \left(\pi \frac{\sqrt[3]{x} - 1}{x - 1} \right)$.

$$\text{sol: } \lim_{x \rightarrow 1} \pi \frac{\sqrt[3]{x} - 1}{x - 1} = \pi \lim_{x \rightarrow 1} \frac{(\sqrt[3]{x} - 1)(\sqrt[3]{x^2} + \sqrt[3]{x} + 1)}{(x - 1)(\sqrt[3]{x^2} + \sqrt[3]{x} + 1)}$$

$$= \pi \lim_{x \rightarrow 1} \frac{x - 1}{(x - 1)(\sqrt[3]{x^2} + \sqrt[3]{x} + 1)} = \pi \lim_{x \rightarrow 1} \frac{1}{\sqrt[3]{x^2} + \sqrt[3]{x} + 1} = \frac{\pi}{3}$$

$$\therefore \sec x \text{ is conti at } x = \frac{\pi}{3} \quad \therefore \lim_{x \rightarrow 1} \sec \left(\pi \frac{\sqrt[3]{x} - 1}{x - 1} \right) = \sec \left(\lim_{x \rightarrow 1} \pi \frac{\sqrt[3]{x} - 1}{x - 1} \right) \\ = \sec \left(\frac{\pi}{3} \right) = 2.$$

3.5 Properties of Continuous Functions: Intermediate Value**Theorem:**

The Intermediate Value Theorem Suppose that $f(x)$ is continuous on the closed interval $[a, b]$ and let N be any number between $f(a)$ and $f(b)$, where $f(a) \neq f(b)$. Then there exists a number c in (a, b) such that $f(c) = N$.

Remark:

The Intermediate Value Theorem can be used to locate roots of equations.

Example 5.1

Show that $f(x) = x^2 - 3 + \frac{1}{x}$ has at least two real roots in the interval $[\frac{1}{2}, 2]$.

sol: $f(x)$ is conti on $[\frac{1}{2}, 2]$.

$$f(\frac{1}{2}) = \frac{1}{4} > 0, \quad f(2) = \frac{3}{2} > 0, \quad f(1) = -1 < 0$$

① $f(x)$ is conti on $[\frac{1}{2}, 1]$ and $f(\frac{1}{2}) > 0, f(1) < 0$.

$\Rightarrow$ There is $c_1 \in (\frac{1}{2}, 1)$ s.t. $f(c_1) = 0$ by IVT.

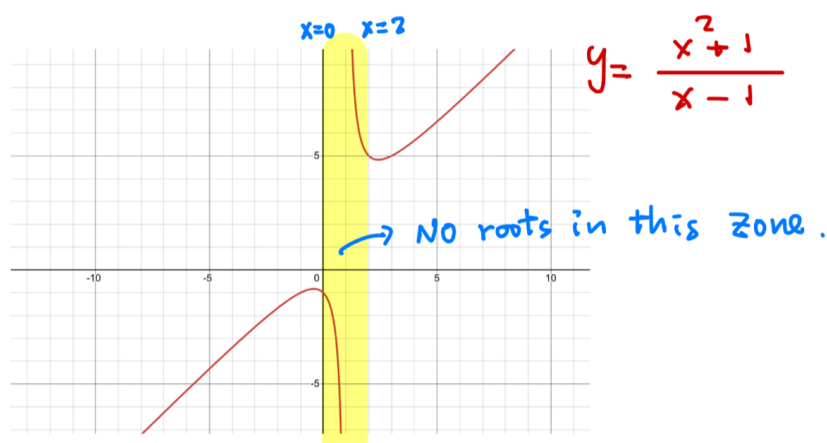
② $f(x)$ is conti on $[1, 2]$ and $f(1) < 0, f(2) > 0$.

$\Rightarrow$ There is $c_2 \in (1, 2)$ s.t. $f(c_2) = 0$ by IVT.

Example 5.2

Does $f(x) = \frac{x^2 + 1}{x - 1}$ have real roots in the interval $(0, 2)$?

sol: $f(0) = -1, f(2) = 5$ But $f(x)$ is NOT conti on $[0, 2]$.
We can NOT apply IVT.

**Remark:**

The Intermediate Value Theorem can be used to prove the existence of fixed points, equilibrium, and other important results. It also serves as a foundation for establishing further properties of continuous functions.

Example 5.3 ①

②

Prove that if $f(x)$ is continuous with domain $[0, 1]$ and range contained in $[0, 1]$ then there is some $c \in [0, 1]$ such that $f(c) = c$. We call c the *fixed point* of $f(x)$.

Sol: Consider $g(x) = f(x) - x$.

$\therefore g(x)$ is the difference of two cont. functions.

$\therefore g(x)$ is cont. on $[0, 1]$.

Moreover, $g(0) = f(0) - 0 \geq 0$, $g(1) = f(1) - 1 \leq 0$.

Case 1 = If $g(0) = 0$ or $g(1) = 0$, then

$f(0) = 0$ or $f(1) = 1$ which implies 0 or 1 is a fixed pt of $f(x)$.

Case 2 : If $g(0) > 0$ and $g(1) < 0$, then by IVT

there is some $c \in (0, 1)$ s.t. $g(c) = f(c) - c = 0$

i.e. c is a fixed pt of $f(x)$.

Exercise 5.1

Suppose that the demand function $D(q)$ and the supply function $S(q)$ are continuous and $D(1) > S(1)$, $D(10) < S(10)$. Show that there is some $q_0 \in (1, 10)$ such that $D(q_0) = S(q_0)$.

Exercise 5.2

Show that a one-to-one continuous function is either increasing or decreasing on its domain. Then, prove that the inverse function of a one-to-one continuous function is also continuous.